

GenAI – Hands on 2

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Problem Title	Daily Horoscope Generator using Text Generation
Problem Statement	People enjoy reading short inspirational messages in horoscopes, but most applications require writing messages manually or using pre-written templates. The goal of this project is to create a system that automatically generates original, generic, but persuasive horoscope messages for any sign of the zodiac. With the help of a text generation model, we can compose personalized messages that seem new every day.
Project Overview	The user inputs a zodiac sign (for example, Leo, Virgo, Scorpio). The program employs a text generation model to produce a 2-3 sentence horoscope that is mystical, encouraging, and open to interpretation, much like actual horoscope applications.
Concepts covered from Unit 1	This project uses the Text Generation pipeline from Hugging Face: pipeline("text-generation") This pipeline enables to predict the next words in a sequence, which allows the model to spin up creative text such as horoscopes.
Hugging Face Model(s) Used	1. gpt2: strong creative generation 2. [OR] distilgpt2: smaller, faster version that works well on laptops.
Input → Process → Output Flow	Example: Input: Cancer Process: 1. Insert the sign (in this case, cancer) into the prompt such that – Daily Horoscope for Cancer 2. Run it through the pipeline("text-generation") with: - temperature: high – for greater creativity - do_sample=True for variation 3. Model generates horoscope-style text (1-3 lines). Output: Horoscope for Cancer
Sample Prompt + Expected Output	Prompt: Daily horoscope for Virgo Generated Output: "Virgo, today brings a sense of quiet clarity that helps you sort through something you've been overthinking. Someone may appreciate your steady guidance more than they admit, giving you a chance to show your calm wisdom. Stay open to small changes — a gentle shift could lead to a pleasant surprise."
Why this project fits Unit 1?	This project fits in unit 1 because it uses Hugging Face pipelines, specifically Text Generation. It showcases: - Prompt design - Pipeline usage - Sampling parameters

	• Model creativity It also generates visible output and also very beginner-friendly for understanding and applying the concepts better.
Limitations	• GPT-2 can sometimes produce text that goes on a tangent or is too long. • It sometimes repeats phrases or produces sentences that are extremely generic. • The model does not actually know any astrology facts — it is pattern-based, not fact-based. • Horoscope tone can vary unless the prompt is carefully controlled.
Planned Final Deliverable	• For the final project, the project will be a small app or script that: • Accepts a zodiac sign as input • Processes a text generation pipeline • Outputs the horoscope in a clean UI or console print-out

Code and Output:

```
import tkinter as tk

from tkinter import font

from transformers import pipeline

# Load model once

generator = pipeline("text-generation", model="distilgpt2")

def generate_horoscope():
    sign = entry.get().capitalize()

    prompt = f"Check and return daily/today's horoscope for zodiac sign {sign}."

    result = generator(
        prompt,
        max_new_tokens=60,
        temperature=0.9,
        do_sample=True,
        top_p=0.95,
        num_return_sequences=1,
        truncation=False
    )
```

```

    )

    text = result[0]["generated_text"]

    horoscope = text.replace(prompt, "").strip()

    # Basic filter for bad output
    if "true-to" in horoscope or len(horoscope.split()) < 5:
        horoscope = "The stars are quiet today, but trust your instincts and stay open to small surprises."

    output_label.config(
        text=f"{sign} Horoscope:\n\n{horoscope}",
        fg="#333333"
    )

# UI setup
root = tk.Tk()
root.title(" ✨ Horoscope Generator ✨ ")
root.geometry("400x300")
root.configure(bg="#f0f0f5")

# Fonts
title_font = font.Font(family="Helvetica", size=16, weight="bold")
label_font = font.Font(family="Helvetica", size=12)

# Title
title_label = tk.Label(root, text="Daily Horoscope", font=title_font, bg="#f0f0f5", fg="#4b0082")
title_label.pack(pady=10)

# Input frame
frame = tk.Frame(root, bg="#f0f0f5")
frame.pack(pady=10)

```

```
tk.Label(frame, text="Enter Zodiac Sign:", font=label_font, bg="#f0f0f5").grid(row=0, column=0,
padx=5)

entry = tk.Entry(frame, font=label_font, width=15, justify="center")

entry.grid(row=0, column=1, padx=5)


# Button

generate_btn = tk.Button(root, text="🔮 Generate Horoscope", font=label_font, bg="#4b0082",
fg="white",

                        command=generate_horoscope)

generate_btn.pack(pady=10)


# Output

output_label = tk.Label(root, text="", wraplength=350, justify="left", font=label_font, bg="#f0f0f5")

output_label.pack(pady=15)


root.mainloop()
```

